

CHE 201 Fall 2019 Midterm 2

Frank Godawski

TOTAL POINTS

90 / 100

QUESTION 1

1 25 pts

1.1 A 18 / 21

- 0 pts Correct
- 3 pts first is wrong
- 3 pts second is wrong
- 3 pts third is wrong
- 3 pts fourth is wrong
- 3 pts fifth is wrong
- ✓ - 3 pts sixth is wrong
- 3 pts seventh is wrong

1.2 B 4 / 4

- ✓ - 0 pts Correct
- 2 pts explanation is not right
- 4 pts Incorrect

QUESTION 2

2 35 pts

2.1 a 10 / 10

- ✓ - 0 pts Correct
- 2 pts W is not labeled
- 2 pts System boundary is not labeled
- 1 pts Qh is not labeled
- 1 pts Qc is not labeled
- 4 pts Qh and Qc are in wrong direction

2.2 b 5 / 5

- ✓ - 0 pts Correct
- 5 pts Wrong or not done
- 2 pts calculation error
- 2 pts Qh is wrong
- 1 pts W is not calculated

2.3 c 5 / 5

- ✓ - 0 pts Correct
- 5 pts wrong or not done
- 2 pts calculation error

2.4 d 5 / 5

- ✓ - 0 pts Correct
- 0 pts max COP is not calculated
- 2 pts calculation error, T not converted to Kelvin
- 5 pts wrong or not done
- 2 pts COP calculated but not W

2.5 e 3 / 5

- 0 pts Correct
- 5 pts wrong or not done
- ✓ - 2 pts not using heat pump is wrong or not done
- 1 pts calculation error

2.6 f 0 / 5

- 0 pts Correct
- ✓ - 5 pts not calculated or wrong
- 1 pts formula is wrong, should be $Q_c/T_c - Q_h/T_h = -\sigma$
- 2 pts formula is wrong
- 1 pts calculation error
- 1 pts Temperatures are not in K

QUESTION 3

3 40 pts

3.1 a 6 / 6

- ✓ - 0 pts Correct
- 3 pts turbine not shown
- 1 pts system boundary not labeled or labeled wrong
- 6 pts not done or wrong

- **1 pts** turbine inlet outlet not shown

3.2 b 5 / 5

- ✓ - **0 pts** Correct
- **2 pts** control volume not listed
- **5 pts** wrong or not done

3.3 C 4 / 4

- ✓ - **0 pts** Correct
- **4 pts** wrong or not done

3.4 d 8 / 8

- ✓ - **0 pts** Correct
- **5 pts** specific volume is wrong
- **1 pts** calculation error, R unit is not right
- **8 pts** not done or wrong
- **3 pts** mass flow rate is wrong

3.5 e 10 / 10

- ✓ - **0 pts** Correct
- **2 pts** Q is not converted to kW, or sign is wrong , should be negative
- **2 pts** h1 is incorrect
- **2 pts** h2 is incorrect
- **10 pts** not done
- **2 pts** pump work not calculated
- **4 pts** mass balance is wrong

3.6 f 7 / 7

- ✓ - **0 pts** Correct
- **2 pts** v1 is wrong or not calculated
- **1 pts** v2 is wrong or not calculated
- **2 pts** calculation errors
- **7 pts** not done or wrong

NAME: Frank Godawski

Please fill in answers in the space supplied and show all your work to receive full credit.
Useful Tables at the end of the Exam.

1. (21 points)

A) State whether the following statements are **True or False** (3 pts each)

F A pump does work on a ^{liquid} gas to increase the pressure and/or elevation.

T Transient operation denotes a change in state with time.

F As velocity decreases in a diffuser, pressure decreases. ^{increases}

T For a closed system, entropy may be produced within the system.

T The maximum coefficient of performance of any refrigeration cycle operating between cold and hot reservoirs at 40°C and 80°C, respectively, is closely 8.

F Entropy production, σ , is path-dependent. $\text{COP}_{\text{max}} = \frac{T_c}{T_h - T_c} = \frac{313}{353 - 313} = 7.825$

F The second Carnot corollary states that all power cycles operating between the same two thermal reservoirs have the same thermal efficiency.

^{max}

B) (4 pts) An inventor suggests an improvement on the power plant cycle. Aware that one half of the heat transferred to the steam in the furnace is discarded in the condenser to the environment, the inventor suggests getting rid of this wasteful component and sending the steam to the pump as soon as it leaves the turbine (cycle contains boiler, turbine and pump). Do you think this new design will work? Explain.

No, the new design will not work with 100% efficiency because, according to the Kelvin-Planck statement, All Q cannot be converted into W .

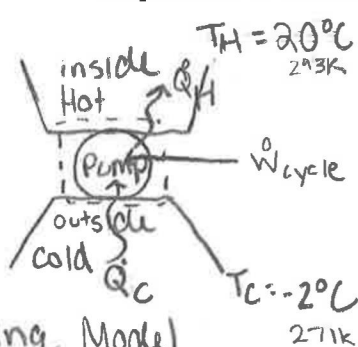
NAME: Frank Giodawski

CHE 201 MidTerm Exam 2

November 22nd, 2019

2. (35 pts) A heat pump is used to meet the heating requirements of a house and maintain it at 20°C in winter. On a day when the average outdoor temperature drops -2°C, the house is estimated to lose heat at a rate of 80,000 kJ/h. If the heat pump under these conditions has a COP of 2.5,

- (10 pts) Draw a schematic showing the heat pump cycle, the thermal reservoirs, T_H , T_C , $\dot{W}_{cycle}$, $\dot{Q}_C$ and $\dot{Q}_H$. Show the direction of the energy transfer rates with arrows.
- (5 points) Determine the power consumed by the heat pump. $\dot{W}_{cycle}$
- (5 points) Determine the rate at which heat is absorbed from the cold outdoor air. $\dot{Q}_C$
- (5 points) Determine the minimum power required to drive this heat pump.
- (5 points) Determine the total heating cost per day (12 cents/ kWh of electricity) of using heat pump. If we were to use electric resistance heater instead of heat pump, how much the cost of heating will be?
- (5 points) Determine the entropy production, σ and state if the cycle is possible, impossible or reversible. Support your answer for full credit.



Engineering Model

- steady state
- inside is hot reservoir
- outside is cold reservoir
- Control volume around pump
- $\dot{Q}$ is positive in the direction of the arrows

b.) $COP = 2.5 = \frac{\dot{Q}_H}{\dot{W}_{cycle}} = \frac{\dot{Q}_H}{\dot{Q}_H - \dot{Q}_C} = 1 - \frac{\dot{Q}_C}{\dot{Q}_H}$

$2.5 = \frac{80,000 \text{ kJ/h} \left(\frac{1 \text{ hr}}{60 \text{ min}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right)}{\dot{W}_{cycle}}$

$2.5 = \frac{22.22 \text{ kW}}{\dot{W}_{cycle}}$

$\dot{W}_{cycle} = 8.889 \text{ kW}$

c.) $\dot{W}_{cycle} = \dot{Q}_H - \dot{Q}_C$

$\dot{Q}_C = \dot{Q}_H - \dot{W}_{cycle}$

$\dot{Q}_C = 22.22 \text{ kW} - 8.889 \text{ kW}$

$\dot{Q}_C = 13.333 \text{ kW}$

d.) $COP_{max} = \frac{T_H}{T_H - T_C} = \frac{293 \text{ K}}{293 - 271 \text{ K}} = 13.318 = \frac{22.22 \text{ kW}}{\dot{W}_{cycle, min}}$

$\dot{W}_{cycle, min} = 1.669 \text{ kW}$

e.) $\left(\frac{0.12}{\text{kWh}} \right) (8.889 \text{ kW}) \left(\frac{24 \text{ hr}}{1 \text{ day}} \right) = \$25.6/\text{day}$

f.) $\sigma = \frac{\dot{Q}_H}{T_C} - \frac{\dot{Q}_C}{T_H} = 0.156 \quad \sigma > 0 \text{ irreversible}$

NAME: _____

CHE 201 MidTerm Exam 2

November 22nd, 2019

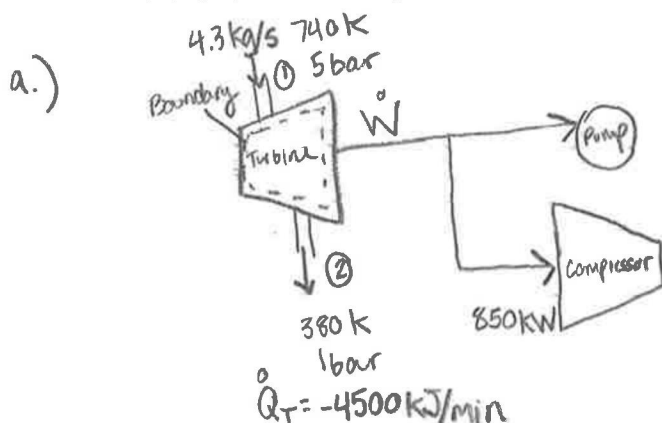
NAME: Frank Grodowski

CHE 201 MidTerm Exam 2

November 22nd, 2019

3. (40 points) An air turbine operating at steady state is providing power to both a pump and an air compressor. The mass flow rate of the air at the turbine inlet is 4.3 kg/s at 740K and 5 bar. The air exits the turbine at 380K and 1 bar. The turbine provides power at a rate of 850 kW to the compressor and the rate of heat transfer from the turbine to its surroundings is 4500 kJ/min. Air can be modeled as an ideal gas and kinetic and potential energy changes are negligible.

- (6 pts) Draw a schematic of the system and label the system boundary.
- (5 pts) State the engineering model
- (4 pts) Write the simplified energy rate balance
- (8 pts) Determine the volumetric flowrate of the air at the turbine exit, in m³/s.
- (10 pts) Determine the power to the pump in kW.
- (7 pts) Sketch the process from State 1 to State 2 on a P-v diagram.



- b.) Engineering Model
- steady state
 - Air is ideal gas
 - $\Delta KE = \Delta PE = 0$
 - Control volume for turbine
 - one inlet/one outlet

c.) Energy Balance (Turbine)

$$\frac{dE_{cv}}{dt} = 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}[(h_1 - h_2)]$$

$$\dot{W}_{cv} = \dot{Q}_{cv} + \dot{m}[(h_1 - h_2)]$$

d.) $\dot{m} = \frac{\dot{V}A}{v_2}$

$$\dot{V}A = \dot{m}v_2$$

Volumetric flow rate

$$P_2 v_2 = RT_2$$

$$v_2 = \frac{RT_2}{P_2} = \frac{(8.314 \text{ kJ/kmol K}) (380 \text{ K})}{(28.97 \text{ kg/kmol}) (1 \times 10^5 \text{ Pa})}$$

$$= 1.0905 \text{ m}^3/\text{kg}$$

$$\dot{V}A = (4.3 \text{ kg/s})(1.0905 \text{ m}^3/\text{kg})$$

$$= \boxed{4.6894 \text{ m}^3/\text{s}}$$

NAME: Franz Gradawski

CHE 201 MidTerm Exam 2

November 22nd, 2019

e.) $\dot{W}_{cv} = \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$

$$h_1 = 756.44 \text{ kJ/kg}$$

$$h_2 = 380.77 \text{ kJ/kg}$$

$$\dot{W}_{cv} = -4500 \text{ kJ/min} \left(\frac{1 \text{ min}}{60 \text{ s}} \right) + (4.3 \text{ kg/s}) (756.44 \text{ kJ/kg} - 380.77 \text{ kJ/kg})$$

$$\dot{W}_{cv} = -75 \text{ kW} + 1615.381 \text{ kW}$$

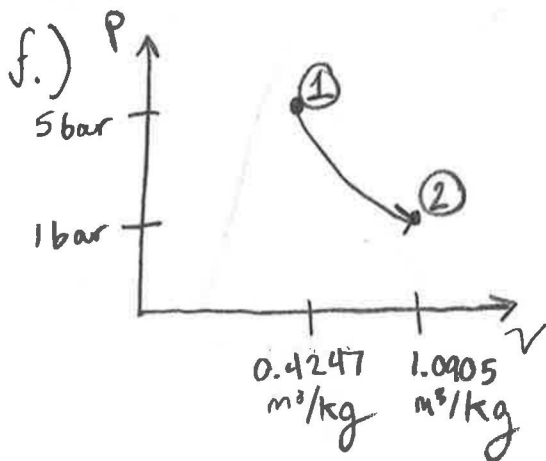
$$\dot{W}_{cv} = 1540.381 \text{ kW} \quad (\text{work from turbine})$$

$$\dot{W}_T = -(\dot{W}_p + \dot{W}_c)$$

$$\dot{W}_p = -\dot{W}_T - \dot{W}_c$$

$$= -1540.381 \text{ kW} - (-850 \text{ kW})$$

$$\dot{W}_p = -690.381 \text{ kW}$$



$$v_2 = 1.0905 \text{ m}^3/\text{kg}$$

$$v_1 = \frac{RT_1}{P_1} = \frac{(8.314 \text{ kJ/kmol}\cdot\text{K}) (740 \text{ K})}{(28.97 \text{ kg/kmol}) (5 \times 10^2 \text{ kPa})} = 0.4247 \text{ m}^3/\text{kg}$$

